OASIS® TRACER

USER MANUAL

PLA PRINT SETTINGS — 0.4mm NOZZLE

These settings are recommended for best results when printing OASIS Tracer models.

FIRST LAYER HEIGHT COMPENSATION:
+0.01 – +0.02 mm
Can be adjusted to compensate for the build plate unevenness.

RETRACTION SPEED:
MAX 25 – 30 mm/s
To avoid nozzle clogging.

CHAMBER TEMPERATURE:
BELOW 36°C
To avoid filament softening and tangle.

NOZZLE TEMPERATURE:
215 – 220°C
Higher temperature is needed to compensate, since the next layer can take time to apply.

FIRST LAYER SPEED:
50 – 70 mm/s

REST OF THE PRINT SPEED:
120 – 160 mm/s

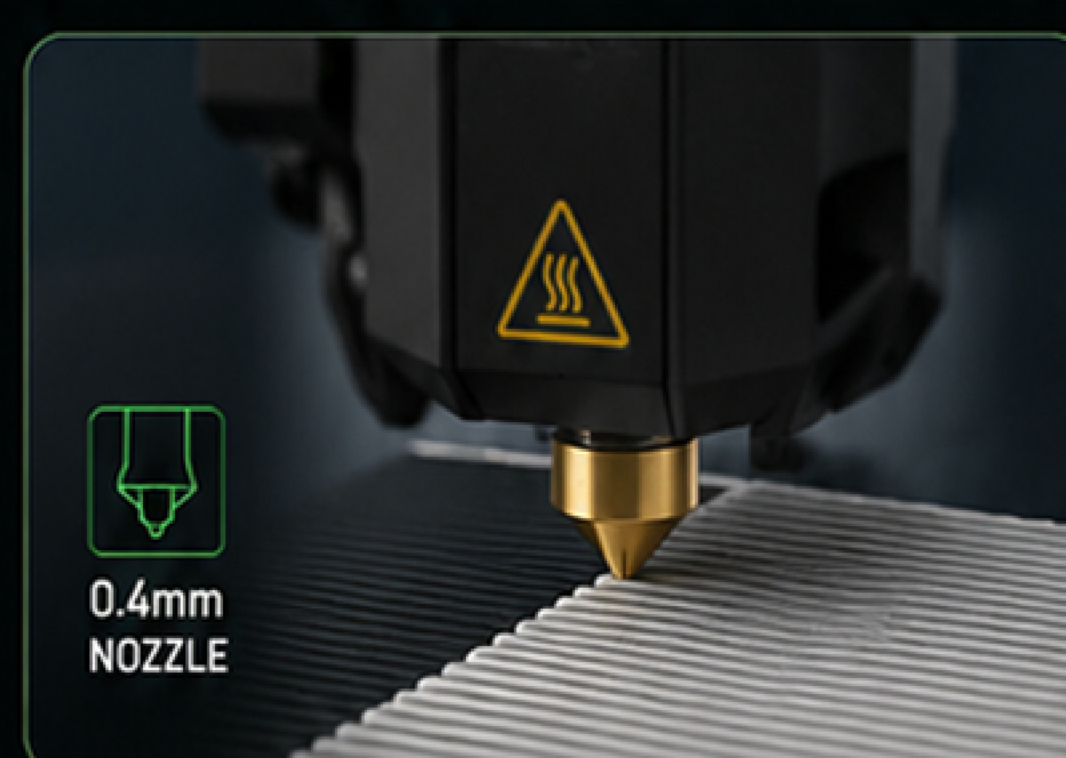
Z HOP:
MANDATORY
Helps prevent surface collisions and artifacts.

BEST BUILD PLATE RESULTS:
TEXTURED PEI
Face down print recommended.

WALL/INFILL OVERLAP:
AT LEAST 20%
Ensures better bonding between walls and infill.

BEST RESULTS WITH:
+2% – +4% FLOW
For stronger extrusion and better layer bonding.

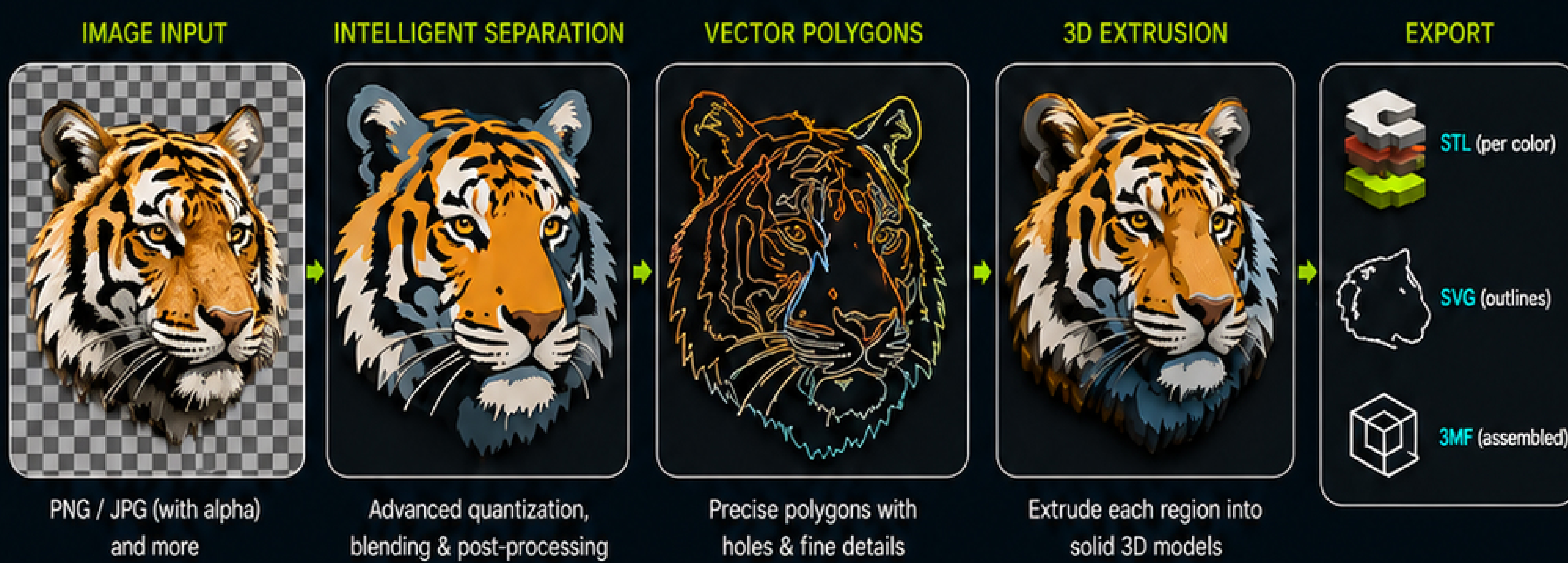
BEST SOLID INFILL RESULTS:
MONOTONIC LINE
Provides the strongest and cleanest surfaces.



Because OASIS models use thin layered geometry, stable extrusion and controlled thermal behavior are critical for best visual quality and layer adhesion.

OASIS TRACER

VECTORIZATION. EXTRUSION. EXPORT.



KonseptID Studio

Oasis family

POWERFUL FEATURES

LOADS COMMON IMAGE FORMATS

Loads common image formats (PNG, JPG with alpha transparency), and other formats.

INTELLIGENT SEPARATION

Intelligently separates the image into clean color regions using advanced quantization, blending, and post-processing.

PRECISE VECTOR POLYGONS

Converts regions into precise vector polygons while preserving holes and fine details.

3D EXTRUSION

Extrudes each region into solid 3D models with customizable thickness.

MULTIPLE EXPORT OPTIONS

Exports one clean STL file per color or blend, plus SVG outlines for laser cutters and CNC, and fully assembled 3MF files.

STABLE & RELIABLE STL MODELS

The program is optimized for creating stable STL models. Even if non-manifold, the models are constrained in their own zone and do not affect the quality or the slicing.

FAST & EFFICIENT

Optimized for low resource usage and fast generation of stable models.

OASIS TRACER v1.02 – EXPORT SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 STL LABELS

PURPOSE

Controls export of individual STL files for generated model regions.

HOW IT WORKS

When enabled, Oasis Tracer exports separate STL models for generated components.

Examples:

- color regions
- blend modifier parts
- filler strip
- raft components
- lightbox walls

These STL files can be:

- imported manually
- packed into a 3MF
- edited individually
- assigned colors manually in slicers

IMPORTANT

STL Labels can be disabled if you do not want individual STL export when generating with Blend Modifiers.

Useful for:

- cleaner output folders
- lower STL count
- simplified workflow

DEFAULT

STL Labels: **Enabled**

Recommended for most workflows.

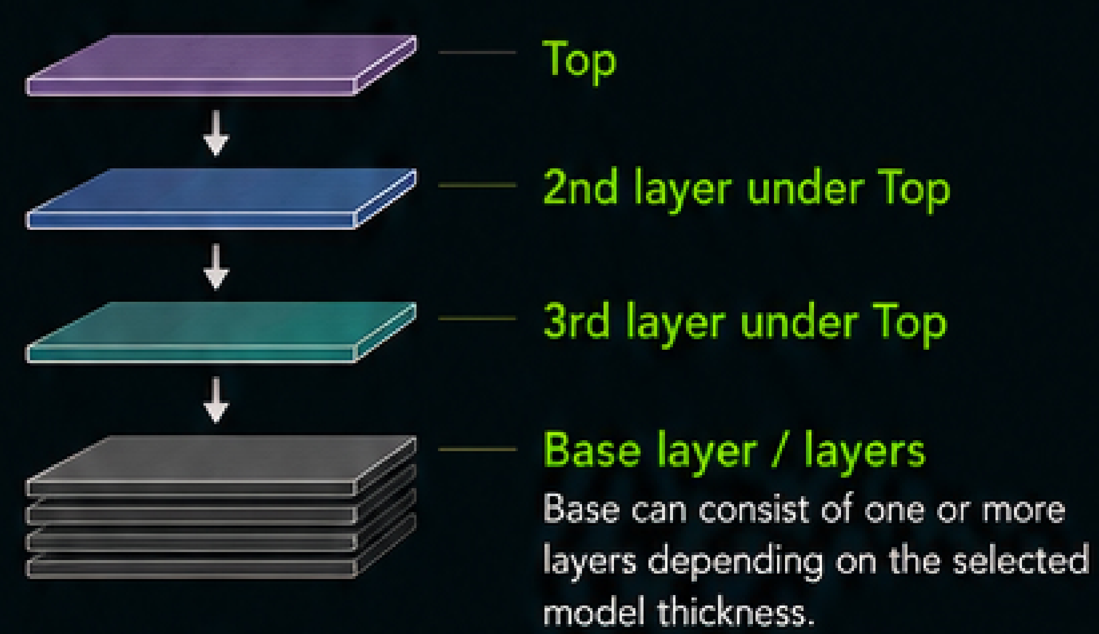
2 BLEND MODIFIERS

PURPOSE

Creates additional modifier geometry to simulate smoother physical color transitions after printing.

HOW IT WORKS

Blend modifiers are generated from upper layers using a Top → Down approach:



Blend modifiers are generated only from the 2nd and 3rd layers below the Top layer.

These modifier layers create intermediate color transitions and improve visual blending after printing.

Example:

Without Blend Modifiers	With Blend Modifiers
Blue → Purple	Smoother transition
Blue → Purple	Blue → DarkBlue modifier → Red modifier → Purple appearance

RECOMMENDED USAGE

Ultra small details with large count	No blends recommended. There may not be enough available geometry tolerance for moving, splitting and modifier generation.
Complex images with small details	**Blend Layers: Top 1** Preserves detail while allowing limited blending.
General use	**Blend Layers: Top 2** Recommended default. Provides balanced blending and good detail preservation.

ADDITIONAL NOTES

Blend modifiers:

- increase STL count
- increase processing time
- increase slicing time
- improve color transitions
- improve visual smoothness

DEFAULT

Blend modifiers: **Top 2**

Safe default for most images.

IMPORTANT LAYER HEIGHT NOTE

When using Blend Modifiers, manually calculate:

- model thickness
- raft height
- lightbox wall height (if used)

Blend modifiers operate using fixed 0.08 mm layer steps.

0.08 mm
0.16 mm
0.24 mm
0.32 mm
...

Changing model height incorrectly, or using different slicer layer heights, can alter layer alignment and negatively affect color transitions.

This may:

- weaken blend quality
- change modifier behavior
- reduce intended color mixing
- produce incorrect visual results

For best results:

Keep Blend Modifier structures aligned to 0.08 mm steps.

Recommended examples:

Base + Top1 + Top2 + Top
= 4 × 0.08 = 0.32 mm

Base + Top1 + Top
= 3 × 0.08 = 0.24 mm

OASIS TRACER v1.02 – ADVANCED SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 BLEND MIN WIDTH

Controls the minimum printable width allowed for Blend Modifier generation.

PURPOSE

Blend Modifiers need physical space to split, move and create smooth color transitions.

When a region becomes thinner than the selected Blend Min Width, modifier generation is restricted or simplified to preserve clean, printable geometry.

HOW IT WORKS

The program analyzes available geometry before creating blends. Only regions that are wider than the selected value are used for modifier creation.

Higher values:

- reduce tiny and noisy blends
- improve print stability
- improve usability on complex images
- reduce excessive small modifier regions

Lower values:

- preserve very small blend details
- allow more modifier generation
- create more aggressive color transitions

RECOMMENDED VALUES

0.2 mm — Very small sharp details.
Required for:
- text
- logos
- line art
- highly detailed geometry

0.4 mm — Complex images with many colors and fine details.

0.6 mm — Recommended default for most images.
Provides better stability and cleaner modifier generation.

DEFAULT

Blend Min Width: 0.6 mm

Recommended for most images that do not contain excessive tiny details or image noise.

SCALE BEHAVIOR

Blend Min Width automatically adapts to the final model size. Changing the Target Width (print width) changes how the same value behaves in the final model.

Smaller print size	Balanced	Larger print size
Less physical space Same value behaves more aggressively	Default target width 230 mm	More physical space Same value behaves more gently

Smaller models compress geometry,
so the same Blend Min Width value affects more areas.
Larger models provide more room for clean blends.

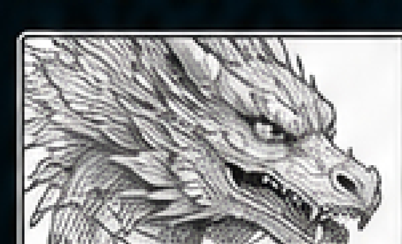
EXAMPLE COMPARISON

The same Blend Min Width value behaves differently depending on the Target Width.

Target Width	px/mm	Example Setting	How it behaves
230 mm (Default)	≈ 21.74 px/mm	0.6 mm	Balanced. Good default behavior.
190 – 200 mm	≈ 25 – 26.3 px/mm	0.8 mm	Behaves similarly to ~0.4 – 0.5 mm at larger print sizes.
300 mm	≈ 16.67 px/mm	0.6 mm	Behaves more gently. More details preserved.

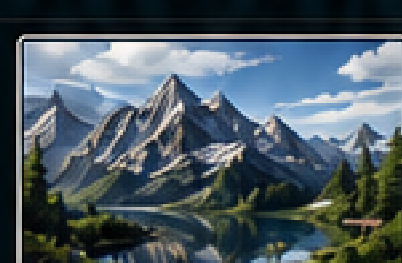
There is no need to recalculate manually.
Simply choose a value based on the image detail and let the program adapt to your target width.

PRACTICAL GUIDELINES



Very small details / sharp lines
Use 0.2 mm
Keeps thin features clean and printable.

Many colors / high detail
Use 0.4 mm
Better usability and fewer tiny, unstable modifiers.



General use / most images
Use 0.6 mm
Better stability, cleaner geometry and smoother results.

NOTES

- Too low values may create many tiny modifier parts that are difficult or impossible to print.
- Too high values may remove important blend details and create visible color steps.
- Adjust mainly based on the image detail and the amount of small features, not just the number of colors.

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

3 SLICER SETTINGS & COLOR MANAGEMENT

Follow these recommendations for best results when preparing and printing your models.

Correct slicer settings and color management ensure accurate colors, clean surfaces and strong, reliable prints.

1. RUNTIME COLOR PALETTE (OASIS COLOR LIST)

Oasis uses a runtime color palette that you can customize with your favorite colors.

Add colors with their RGB values so the program can read and apply them.

1. Open installation folder (Oasis Tracer)\runtime\colors\
2. Open the palette.txt file (not json)
3. Add vendor colors with RGB values
4. Save the file. Restart Oasis Tracer.

palette.txt

```
# Oasis Tracer Color Palette
# Name,R,G,B
Prusa Orange,255,106,0
Prusa Galaxy Black,13,17,23
Prusa Arctic White,255,255,255
Bambu Jade White,222,245,242
Bambu Green,0,179,107
Elegoo Grey,112,122,131
```

EXAMPLE PALETTE ENTRY (palette.txt)

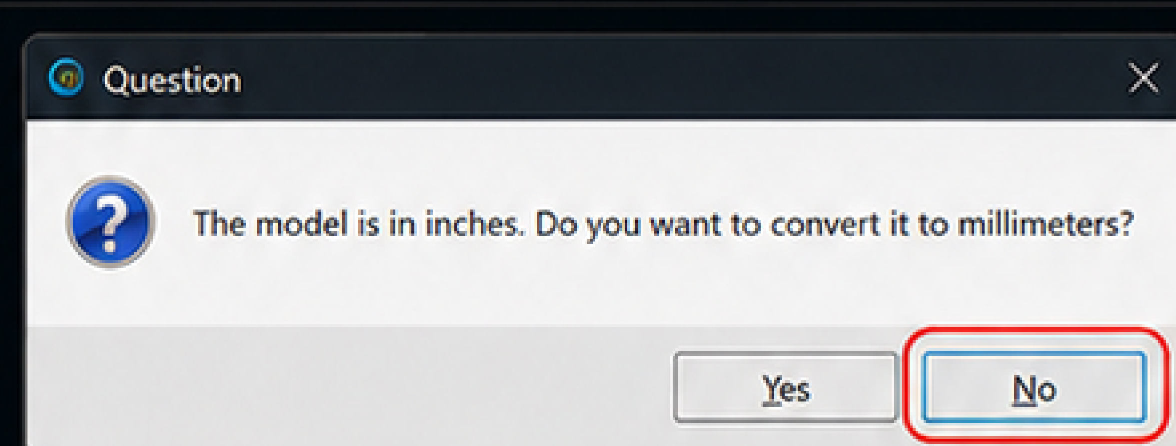
```
# Oasis Tracer Color Palette
# Name,R,G,B
Prusa Orange,255,106,0
Prusa Galaxy Black,13,17,23
Prusa Arctic White,225,255,255
Bambu Jade White,222,245,242
Bambu Green,0,179,107
Elegoo Grey,112,122,131
```

RESULT IN OASIS

- Use RGB values (0-255).
- Give each color it's vendor specific name.
- After saving, the colors will be available in Oasis Tracer.

2. WHEN LOADING STL / 3MF FILES

Some slicers may prompt you to convert the model to millimeters or inches.
Always click NO.
Oasis exports models in millimeters.
Converting can cause incorrect scaling.



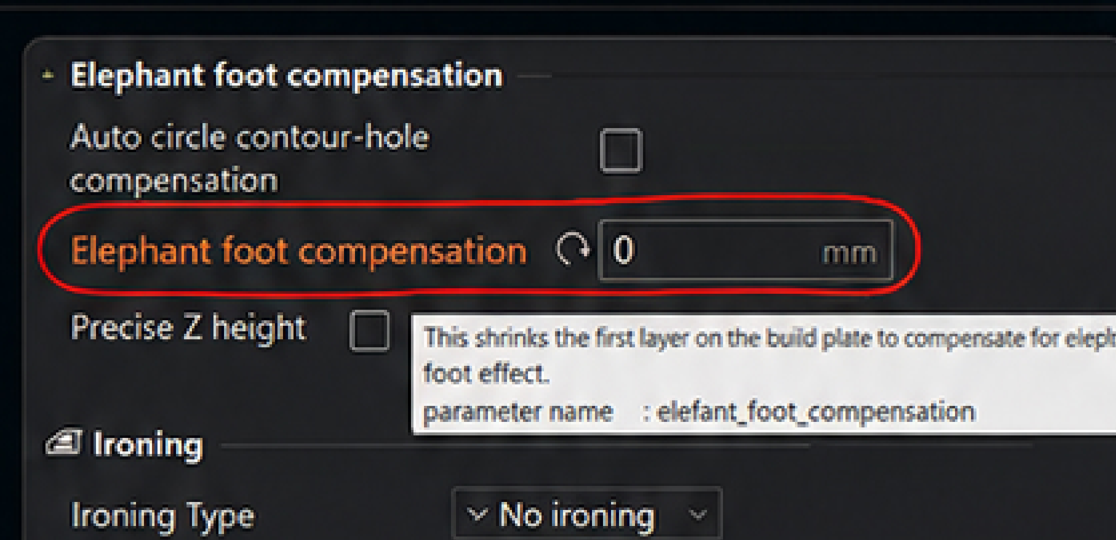
Always click **NO** to keep the original scale.

3. PRINTING MIRRORED BY Z (TO AVOID SUPPORT)

When you mirror the model by Z axis (print upside down) to avoid support and use a light box:

Set Elephant Foot Compensation to 0.

This prevents the first layer from being shrunk and ensures the model sits flat and fits correctly in the light box.



Elephant Foot Compensation must be set to 0 mm.

4. RECOMMENDED PRINT SETTINGS

For the best balance of quality, strength and surface finish, use these recommended settings.

INFILL PATTERN
Use Monotonic line infill pattern.
Provides clean surfaces and consistent strength.

WALL LOOPS
Use 1 wall loop.
This preserves fine details and keeps the model true to the original design.

OTHER RECOMMENDATIONS
- Layer height: 0.12 – 0.20 mm
- Top/bottom layers: 3 – 5
- Slow down for small details

ADDITIONAL TIPS

- Always preview before printing to verify colors and layers.
- Use the same filament brand/type for all colors when possible.
- Enable "Detect Thin Walls" in your slicer for small detail preservation.
- Calibrate your first layer for optimal adhesion and surface quality.

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

2 RELIEF / ACCENT DESIGN

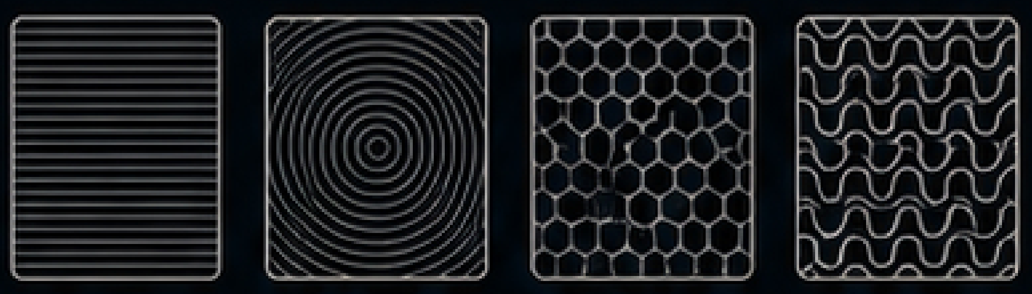
Raise, lower or remove any part to create relief (3D effect) or to hide elements. Perfect for text, logos, shapes and key details.

Each segmented model is independent. You can scale it up or down on the Z axis without affecting other parts.

APPLY DIFFERENT TOP SURFACE PATTERNS

Each model can use its own top surface pattern in the slicer.

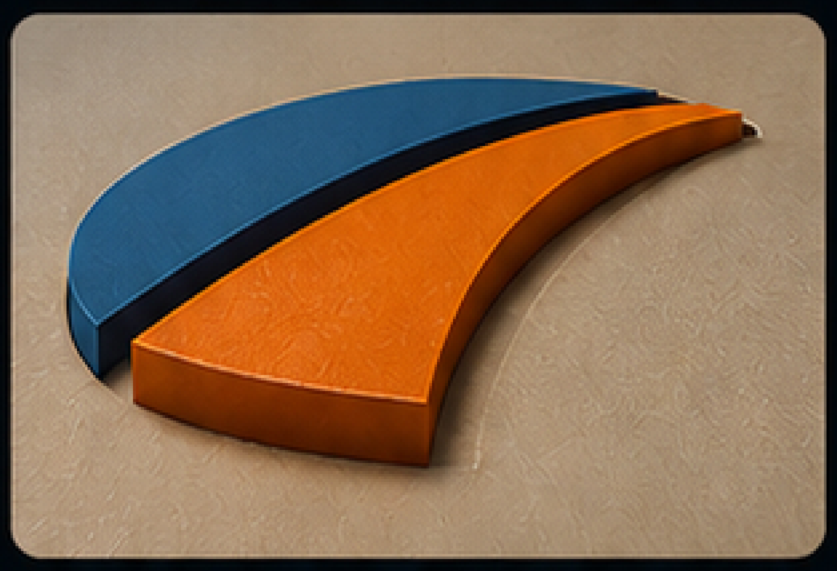
Apply unique textures or finishes to individual parts for more creative and functional results.



Lines | Concentric | Honeycomb | Gyroid | ...and more

MORE RELIEF POSSIBILITIES

RAISE ANY SHAPE OR LOGO



Logos, icons or any shape can be raised to stand out from the background.

LOWER / ENGRAVE ANY SHAPE

Lower a shape below the surface to create engraved or recessed details.

REMOVE ANY SHAPE

Remove a model entirely to create openings or cut-outs. (No fill is added.)

You can hide, raise, lower or remove any segmented model.
Use this to highlight what matters most in your design.

- Raise (Positive Z)
- Lower (Negative Z)
- Remove (Hide)

BEST FOR

- Signage & Advertising
- Logos & Branding
- ABC Text Emphasis
- Decorative Accents
- Layered Designs
- Custom Cut-Outs & Openings

NOTE
Scaling is applied only on the Z axis (height). The original X/Y shape and position remain exactly the same.

IMPORTANT
Removing a shape does not fill the space. The area will be empty (open).

SLICER WORKFLOW EXAMPLE

1. Select Model (Text, Logo, Shape)
2. Scale on Z Axis (Raise / Lower)
3. STL Export STL
4. Slice & Print

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

SEPARATION MODES

These modes control how the image is separated into regions and colors before vector generation.

1 TONE

PURPOSE

Creates separation based on brightness (tone) levels. Ideal for clean, high quality images with clear light and dark areas.

HOW IT WORKS

The image is analyzed by brightness and divided into distinct tone layers.
Color information is ignored. This produces clean, smooth regions with strong contrast separation.

BEST FOR

High quality and vector like images with clean lines and strong contrast.

EXAMPLES

RECOMMENDED USE

- Logos and vector graphics
- Line art and manga
- High contrast illustrations
- Images with minimal gradients

NOTE

Best for high quality and vector like images.

2 HYBRID REGION

HYRPOSE

Balanced separation that combines color and tone analysis. Designed for gradients and multicolor images.

HOW IT WORKS

The image is analyzed using both color similarity and tonal structure.
Regions are created based on a balance between color distance and brightness consistency, preserving smooth gradients while keeping regions clean.

BEST FOR

Gradients and multicolor images with medium to high complexity.

EXAMPLES

RECOMMENDED USE

- Digital paintings
- Gradients and smooth shading
- Multicolor illustrations
- Photos with medium complexity

NOTE

Best for gradients and multicolor complex image separation.

3 K-MEANS

PURPOSE

Advanced color clustering using K-Means algorithm. Provides maximum detail preservation for complex images.

HOW IT WORKS

The image is clustered into the optimal number of colors using the K-Means algorithm.
This creates highly accurate regions and preserves very small details, textures and sharp edges.

BEST FOR

Complex images with very small, sharp details (like texts).

EXAMPLES

RECOMMENDED USE

- Complex images with tiny details
- Images containing text or small symbols
- Photographs with high detail
- Textures and intricate patterns

NOTE

Almost mandatory for complex images with very small, sharp details (like texts).

TIP Not sure which mode to use?

Start with Hybrid Region. If the result lacks detail, try K-Means. If the result is too noisy or has too many small regions, try Tone or increase Final island for stronger cleanup.

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 SELECTIVE STL EXTRACTION

Extract any color or shape you need and use it for other projects.

Oasis segments your image into independent models.
You can export any single model (one color or shape) as STL and use it anywhere.

Oasis exports separated geometry rather than a single combined slab.

This gives you complete freedom to use each part your way.

HOW IT WORKS

1. Segmented Image

- Black
- Dark Gray
- Light Gray
- Red
- Skin
- White

Image is automatically segmented into separate models by color/region.

2. Select One Model

☑ Black (selected)

Enable only the model you want to extract.

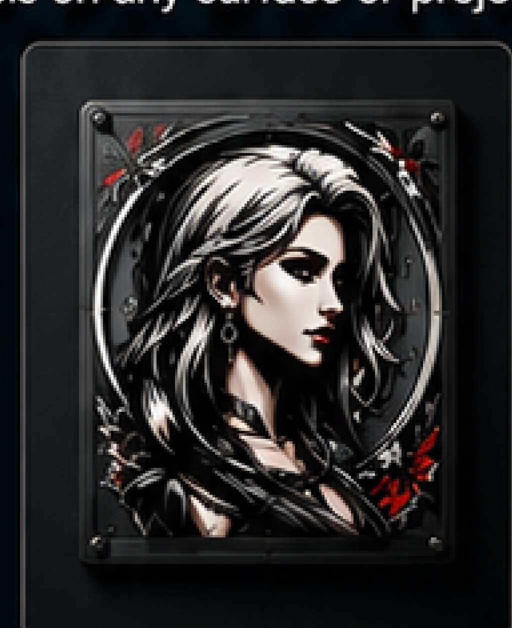
3. Export STL

Export the selected model as STL.

EXAMPLE USES

Use extracted STL models on any surface or project you want.

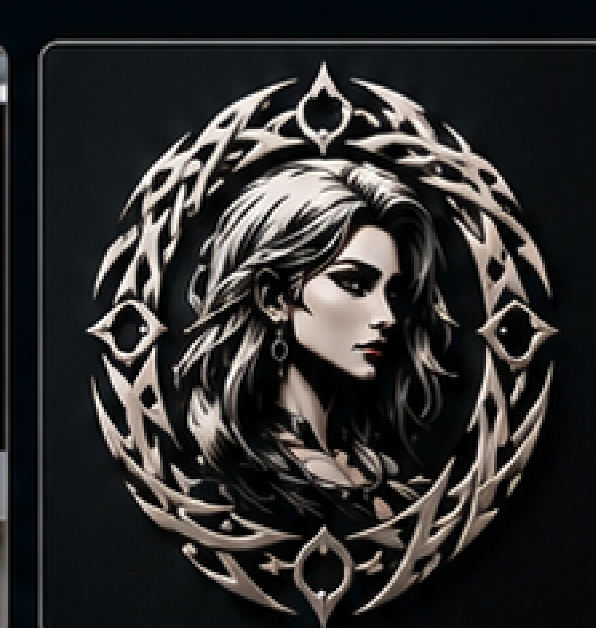
TPU T-Shirt Print
Flexible and durable single-color print.



Logo / Badges
Add to logos, badges or name plates.

Wall Art / Decor
Create unique art and decorations.

Stickers / Decals (SVG Outline)
Export SVG outline for vinyl cutting, stickers, decals and more.



3D Tribal Models
Create 3D tribal shapes in seconds.

You are not required to print all exported models.
Any individual STL can be used independently in other designs, prints or materials.

TIP Need a specific shape or part?
Simply enable the region you want, export it as STL, and use it in your favorite 3D software or slicer.

Extract → Export STL → Use Anywhere

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 THICKNESS mm

PURPOSE

Controls the total vertical thickness of the generated STL model.

HOW IT WORKS

- The Thickness value defines the overall slab height of the exported model.
- In standard modes, this is the direct model thickness.
- In Blend modifiers mode, the thickness represents the combined thickness of all generated layers, including:
  - Base layer
  - Internal blend layers
  - Top layer

Internal blend layers always use a fixed thickness of **0.08 mm** per layer.

Because of this, increasing blend layers changes the required total slab thickness.

EXAMPLE (DEFAULT STACK)

1 Base layer	2 Blend layers	1 Top layer	Total thickness
0.08 mm	2 × 0.08 mm	0.08 mm	4 × 0.08 mm = **0.32 mm**

This is the default configuration and recommended safe starting point.

RECOMMENDED VALUES

0.32 mm
General multicolor work. Balanced color blending, print time and model strength.

0.40 mm
Provides stronger color presence, stronger physical parts and easier handling.

DEFAULT

Thickness: **0.32 mm**
Blend layers: **Top 2**

Equivalent to:
1 Base layer
2 Blend layers
1 Top layer

This default is intended as a safe balance between color blending quality, print stability and model strength.

2 ALPHA CUTOFF

PURPOSE

Controls transparency detection and determines what pixels become part of the model silhouette.

HOW IT WORKS

Pixels with alpha values below the selected threshold are treated as transparent and excluded from generation.

Pixels above the threshold become part of:

- Main model silhouette
- Raft
- Lightbox walls
- Filler strip
- SVG outlines
- STL geometry

This setting directly affects the shape of the generated object.

Lower values preserve faint semi-transparent edges. Higher values remove weak transparency and create cleaner silhouettes.

RECOMMENDED VALUES

Value	Description
1	Recommended default. Preserves nearly all visible image information.
10 – 30	Useful for PNGs with soft transparency or anti-aliased edges.
50+	Aggressive cleanup for noisy images or partially transparent artwork.

EXAMPLES

LOW ALPHA CUTOFF (e.g. 1 – 5)

- Soft shadows preserved
- Faint edges retained
- More detail

HIGH ALPHA CUTOFF (e.g. 50+)

- Clean silhouette
- Small transparent fragments removed
- Sharper outer shape

DEFAULT

Alpha cutoff: **1**

This value is recommended for most PNG and transparent artwork workflows.

GOOD PRACTICE

These two settings have the biggest impact on the final result.
Adjust Thickness for physical strength and Alpha cutoff for silhouette accuracy.

STL MODEL | LIGHTBOX | RAFT | SVG EXPORT

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

3 TARGET WIDTH (PRINT WIDTH)

Controls the final physical width of the generated model.

PURPOSE

Target Width defines the maximum width of the model in millimeters on the print bed.

The image is scaled so that its longest side fits this width.

All other settings (Min Island, Smooth, Simplify, Blend Min Width, etc.) are calculated relative to the final scaled size.

HOW IT WORKS

The program uses your image resolution (px) and scales it to match the Target Width (mm).
A px/mm resolution is calculated and used as the reference for all size-based operations.

$$\text{px/mm resolution} = \frac{\text{image long side (px)}}{\text{target width (mm)}}$$

Example:
A 5000 px long side with Target Width 230 mm
→ 5000 / 230 = 21.74 px/mm
This means 1 mm in the model equals ~21.74 px in the image.

VISUAL EXAMPLE

Image (example)
5000 px (long side)

Generated model
Target Width = 230 mm

The model is scaled so the longest side becomes 230 mm.
All other dimensions are scaled proportionally.

PX/MM REFERENCE

The px/mm value is the foundation for all size-based settings.
Higher Target Width → higher px/mm → more detail and precision.
Lower Target Width → lower px/mm → less detail, faster processing.

Target Width (mm)	Example px/mm	Result
115 mm	43.48 px/mm	Lower detail, smaller model
230 mm (default)	21.74 px/mm	Balanced detail (default)
300 mm	16.67 px/mm	More detail
400 mm	12.50 px/mm	High detail
500 mm	10.00 px/mm	Very high detail

RECOMMENDED VALUES

1 – 230 mm	Small to medium models. Good for most standard printers.
230 – 350 mm	Balanced detail and print size. Recommended for most users.
350 mm+	High detail and maximum quality. Requires a larger print plate.

IMPORTANT NOTES

- A 5000 px image at 500 mm Target Width creates a model over 230 mm.
- All size-based settings (Min Island, Blend Min Width, Final Island, Smooth, etc.) scale with the Target Width.
- Using a higher Target Width gives better results and more detail, especially for large or complex images.
- Make sure your printer has enough build plate size for the chosen Target Width.

DEFAULT

Target Width: 230 mm

- Balanced detail and compatibility for most printers.

PRACTICAL EXAMPLES

Target Width: 150 mm
px/mm: 33.33

- Smaller model
- Lower file size
- Less detail
- Faster processing
- Ideal for small prints

Target Width: 230 mm (Default)
px/mm: 21.74

- Balanced model size
- Good detail
- Balanced file size
- Balanced processing time
- Ideal for most users

Target Width: 400 mm
px/mm: 12.50

- Larger model
- Higher detail
- Larger file size
- Longer processing time
- Ideal for large prints and maximum quality

TIP For the best quality, use the largest Target Width your printer can handle.
More mm = more detail, smoother curves and cleaner final geometry.

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 MIN ISLAND px

PURPOSE

Controls the minimum size of isolated pixel islands that are allowed to remain in the generated model.

HOW IT WORKS

Small disconnected image fragments below this threshold are removed automatically before generation.

This helps remove:

- image noise
- JPG artifacts
- tiny dust-like fragments
- accidental isolated pixels

Only isolated regions are affected.
Connected geometry that belongs to the main silhouette is preserved.

This value is measured in pixels on the normalized working image.

> This affects only isolated islands.
> Parts connected to the main silhouette are never removed by this setting.

RECOMMENDED VALUES

20 – 40	Keeps almost everything. Preserves maximum detail and tiny features.
40 – 80	Balanced cleanup. Removes noise while preserving most important details.
80+	Aggressive cleanup. Removes most isolated fragments and small islands.

EXAMPLES

MIN ISLAND px = 20
(Keeps almost everything)

All tiny islands kept.
Maximum detail preserved.

MIN ISLAND px = 50
(Balanced cleanup)

Noise islands removed.
Details preserved.

MIN ISLAND px = 100
(Aggressive cleanup)

Most isolated fragments removed.
Clean silhouette.

DEFAULT

Min island: **30 px**

The program starts at 30 px.
This value is optimized for most images.

NOTE

Minimum effective value is 20 px.
This is the smallest area that a 0.2 mm nozzle can extrude as a dot/line.

2 SIMPLIFY px

PURPOSE

Controls vector outline simplification before STL generation.

HOW IT WORKS

The algorithm removes small directional changes and unnecessary vertices from generated boundaries.

Higher values:

- fewer vertices
- simpler geometry
- faster STL generation
- faster slicing

Lower values:

- more vertices
- more detail retained
- smoother complex curves

This affects:

- STL geometry
- SVG outlines
- Raft
- Lightbox walls
- Filler strip
- Shared boundaries

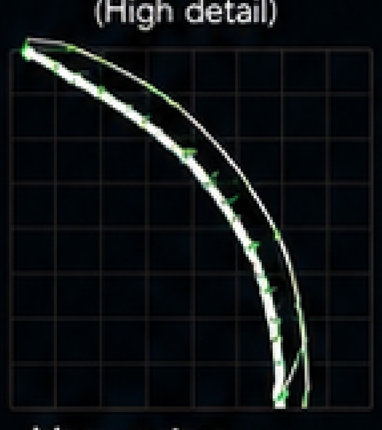
> Higher values = simpler outlines and faster generation, but too high can reduce important shape details.
> Lower values = more detail and smoother curves, but more points in the final geometry.

RECOMMENDED VALUES

1 – 1.5	High detail. Recommended for complex images, anime artwork and detailed curves.
1.5 – 2.5	Balanced mode. Good for most images and general use.
2.5+	Simpler outlines. Not recommended for highly detailed or complex images.

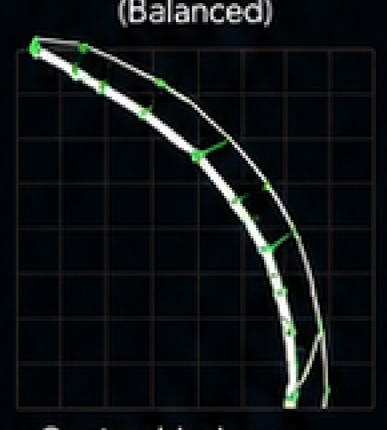
EXAMPLES

SIMPLIFY px = 1.0
(High detail)



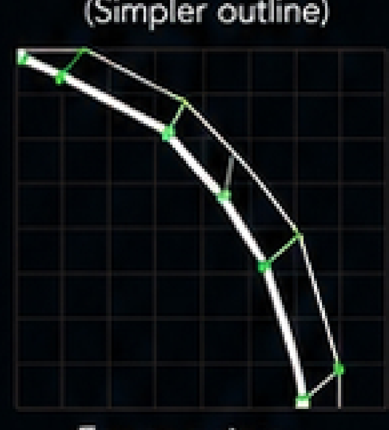
Many points.
Maximum detail and smoothness.

SIMPLIFY px = 2.0
(Balanced)



Optimal balance.
Smooth and clean outline.

SIMPLIFY px = 5.0
(Simpler outline)



Fewer points.
Simpler shape, less detail.

DEFAULT

Simplify: **1.25 px**

Provides high detail while maintaining stable geometry generation.

OASIS TRACER v1.02 – TRACER SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 FINAL ISLAND

PURPOSE

Controls the final cleanup size for isolated regions before STL generation.

HOW IT WORKS

This setting removes small leftover fragments and tiny isolated regions that remain after image processing.

This helps reduce:

- accidental color specks
- tiny isolated fragments
- unwanted small regions
- noisy geometry

Lower values preserve more detail.
Higher values perform stronger cleanup.

RECOMMENDED VALUES

0 – 80	Preserves nearly all details.
80 – 200	Balanced cleanup for most images.
200+	Aggressive cleanup for noisy images and excessive small fragments.

IMAGE RECOMMENDATIONS

0 – 30	Best for images with very small sharp details.
30 – 80	Best for normal quality images.
100+	Recommended for images that require stronger cleanup.

DEFAULT

Final island: **0**

The program starts at 0 to preserve all image information.

2 SMOOTH px

PURPOSE

Controls boundary smoothing before vector generation.

HOW IT WORKS

Smooth px reduces staircase artifacts and local zig-zags along generated boundaries.

This smoothing affects:

- model boundaries
- shared label boundaries
- filler strip
- raft geometry
- lightbox walls
- SVG outlines

Lower values:

- preserve corners
- preserve sharp details

Higher values:

- create softer curves
- reduce sharp pixel transitions
- simplify noisy outlines

RECOMMENDED VALUES

0 – 0.5	High detail. Preserves sharp features.
0.5 – 1.5	Balanced smoothing for most images.
1.5+	Stronger smoothing. May soften very small details.

DEFAULT

Smooth px: **0.5**

Balanced smoothing while preserving detail.

OASIS TRACER v1.02 – EXPORT SETTINGS

These settings define the final geometry, transparency handling and overall model generation.

OASIS TRACER

1 RAFT

PURPOSE

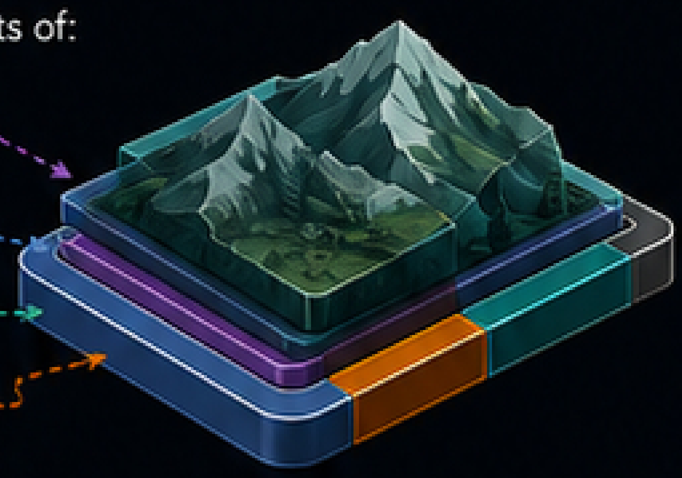
Creates a printable support base below the generated model and provides stability, stacking support and optional visual effects.

HOW IT WORKS

The raft sits directly under the model and creates separate support geometry.

The raft system consists of:

1. Inner raft / base
2. Outer raft strip
3. Inner raft section
4. Filler strip (optional)



Inner raft section and outer raft strip sit next to each other as the filler strip and model, not on top of each other.

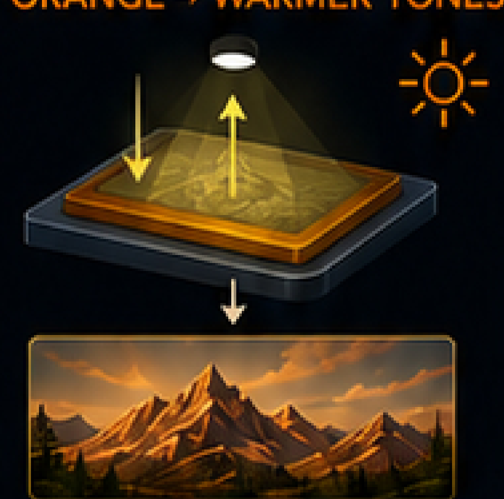
The top surface of the raft becomes Z = 0 for model stacking.

INNER RAFT NOTES

For Lightbox workflows:
White or transparent filament is recommended because it allows better light penetration.

The inner raft can also intentionally use different colors to create color filtering effects:

ORANGE → WARMER TONES



BLUE → COOLER TONES

This can subtly alter the appearance of upper printed layers.

RAFT RECOMMENDED VALUES

LIGHTBOX MODE
0.32 mm
Default and maximum value.

- good light penetration
- enough strength
- reliable support for small details

NORMAL OUTPUT + BLEND MODIFIERS
0.64 mm

- ~1 mm total structure thickness
- stronger parts
- good rigidity
- relatively fast print time

Blend structures use fixed 0.08 mm steps. Calculate raft and model thickness relative to the fixed blend layer stack.

IMPORTANT

When using Blend Modifiers, calculate raft and model thickness relative to the fixed blend layer stack (0.08 mm).
Incorrect heights may reduce intended blending behavior and visual quality.

DEFAULT

Raft: 0.32 mm

Balanced default value.

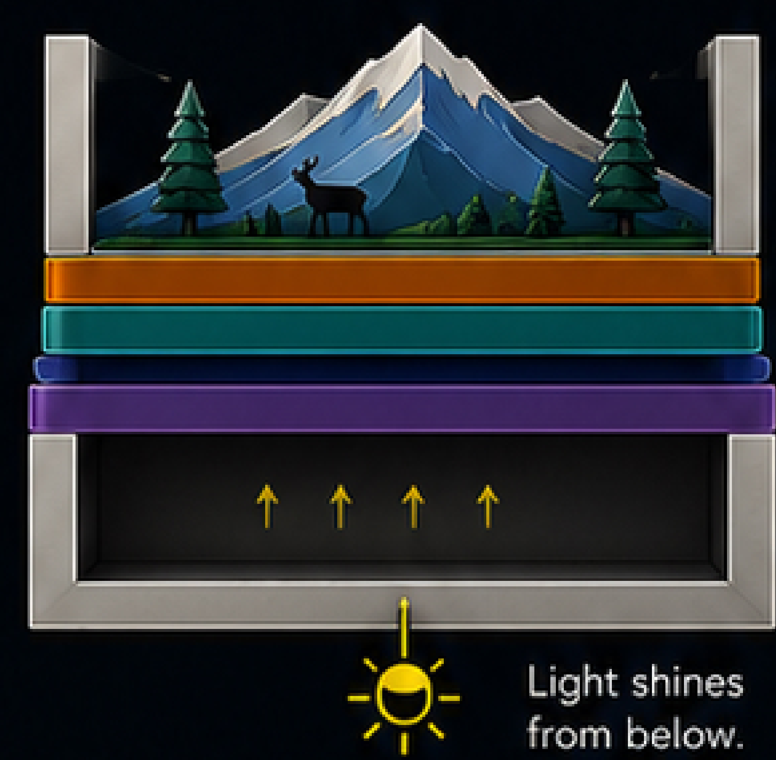
2 LIGHTBOX

PURPOSE

Creates a wall frame under the model to form a lightbox. Useful for illuminated prints, framed artwork and edge containment.

HOW IT WORKS

The lightbox wall sits below the raft and model.
Only the lightbox is flipped around and confusing.



Lightbox walls, filler strips and raft components can use different materials or colors for full creative control.

OFFSET

Controls Lightbox wall thickness.

Lower values
- thinner wall
- lighter structure

Higher values
- thicker wall
- stronger frame

RECOMMENDED
Offset: **1.6 mm**

Balanced wall thickness for most use cases.

WALL HEIGHT

Controls Lightbox wall height.

Lower values
- smaller frame
- shorter print time

Higher values
- deeper frame
- stronger enclosure effect

RECOMMENDED VALUES
20 – 40 mm

Balanced height range for most applications.

DEFAULT

- Lightbox: OFF
- Offset: 1.6 mm
- Wall height: 20 mm

MATERIAL & COLOR FLEXIBILITY

Raft base, outer raft strip, inner raft section, filler strip and lightbox wall can use different colors or materials.

Mix and match for strength, aesthetics and light control.